



POST AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY



Embedded Systems Design

Faculty: Computer science – IT1

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The overall objective of the course is to equip learners with fundamental knowledge and practical skills necessary for designing and developing a dedicated computer system that integrates both hardware and software.

- ❖ The course covers the definition, characteristics, and stringent requirements of embedded systems (ES), such as real-time constraints, reliability, and the ability to operate in harsh environments.
- ❖ It introduces hardware architecture and the principles for selecting appropriate processors, including central processing units (CPUs), microcontrollers (MCUs), and digital signal processors (DSPs).
- ❖ The course also presents software components deployed on embedded systems, including device drivers, middleware, and system software, with a particular focus on real-time operating systems (RTOS) and methods for managing multitasking.
- ❖ Finally, it addresses modern design criteria and methodologies, hardware–software partitioning skills, and co-design trends aimed at optimizing performance, reducing power consumption, and minimizing product cost.

- ❖ Course code: INT1461
- ❖ Number of credits: 3
- ❖ Prerequisites:
 - Computer Architecture
 - C/C++ Programming
- ❖ **Slides will be uploaded to**
https://drive.google.com/drive/folders/1SF7YuTdXhmksl2W3C0CaIE7WT2LVJJ7n?usp=drive_link



- Huỳnh Thúc Cước, Bài giảng Xây dựng hệ thống nhúng, Học viện Công nghệ BC-VT, 2010.
- David, Liam. "Embedded Systems: Architecture, Programming, and Design" by Raj Kamal." PAKISTAN JOURNAL OF LINGUISTICS 5.1 (2023): 108-113.
- Tammy Noergaard, *Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers*, Newnes, 2005.
- Steve Heath, *Embedded Systems Design*, Second Edition, Newnes, 2002.
- Daniel D. Gajski, Frank Vahid, Sanjiv Narayan and Jie Gong, *Specification and Design of Embedded Systems*, Prentice Hall, 1994.

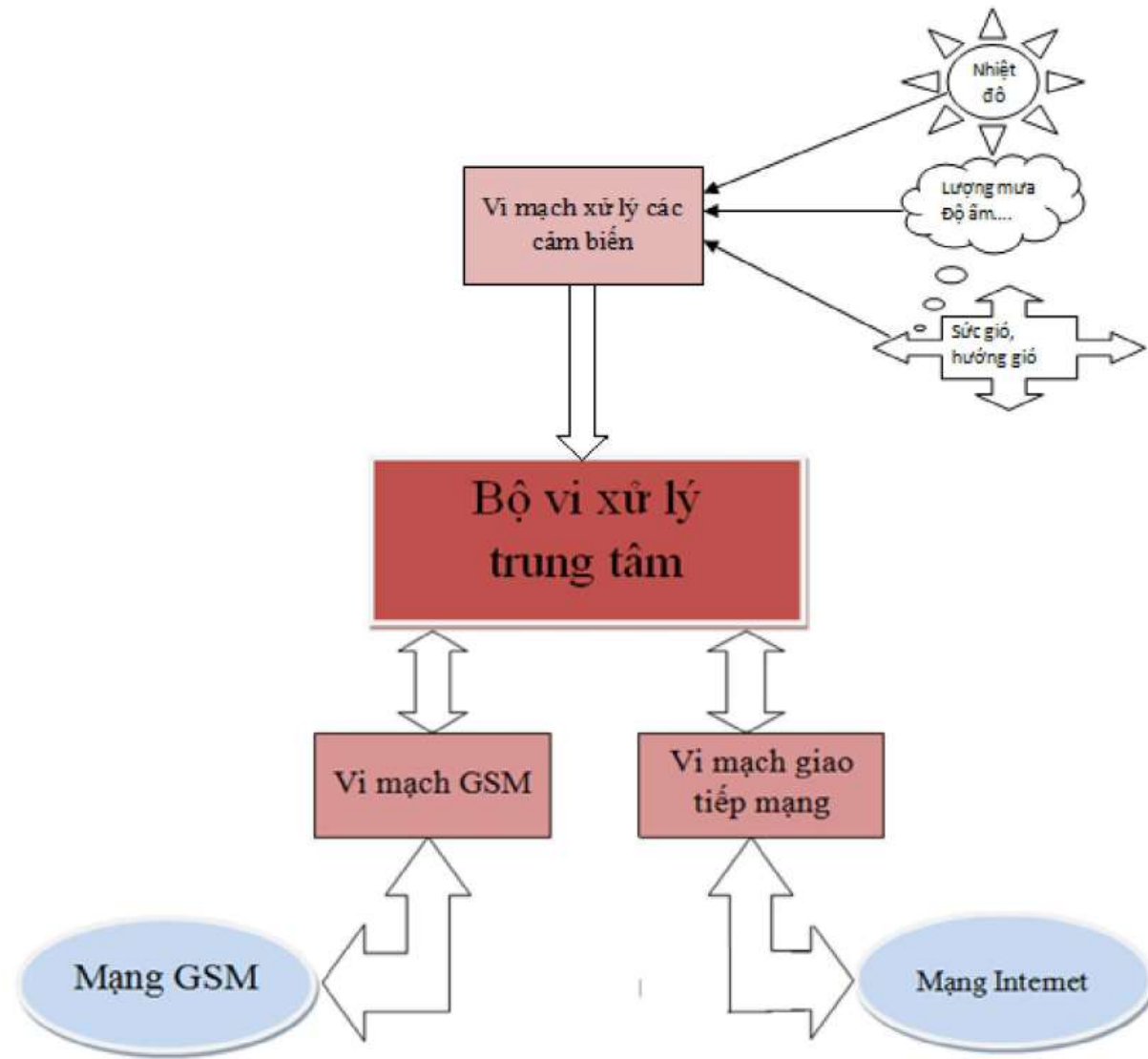
- ❖ Attendance: 10%
 - Attendance is recorded by scanning a QR code at the end of the first session of each class
- ❖ Midterm exam: 10%
 - Individual assessment
 - Conducted in the 6th or 8th class session.
 - Written (essay-based) exam
- ❖ Project: 20%
 - Group work: 4–5 members per group (maximum of 12 groups; registration required before the 3rd class session).
 - Report: To be submitted before the 30th class session (PDF file via email)
 - Presentation: During the final class sessions.
- ❖ Final exam: 60%

Each group is required to complete two reports

- ❖ Additional research assignment: Present an overview of the architecture, features, and applications of a specific processor or microcontroller (excluding the 8051, which is covered in the course).
- ❖ Practical assignment: Use a processor or microcontroller to develop an embedded application
 - Topics are self-selected, with no more than two groups allowed to work on the same topic
 - The project must involve real hardware implementation
 - Examples:
 - Smart temperature and humidity monitoring system
 - Smart alarm clock
 - Mobile-based device control system

Using a microcontroller to simulate SCADA (temperature and water level monitoring)

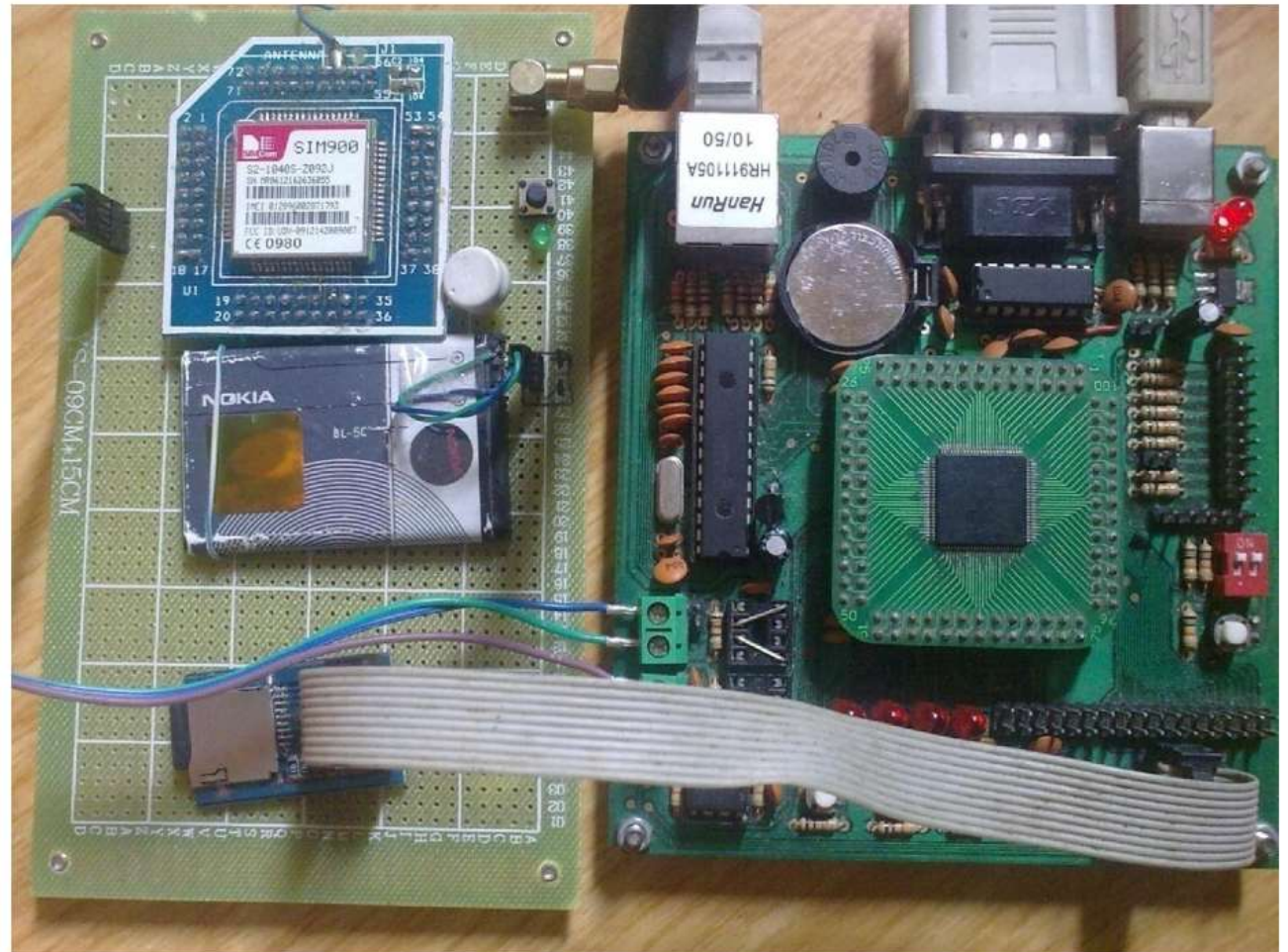
❖ Diagram



Project: Example

Using a microcontroller to simulate SCADA (temperature and water level monitoring)

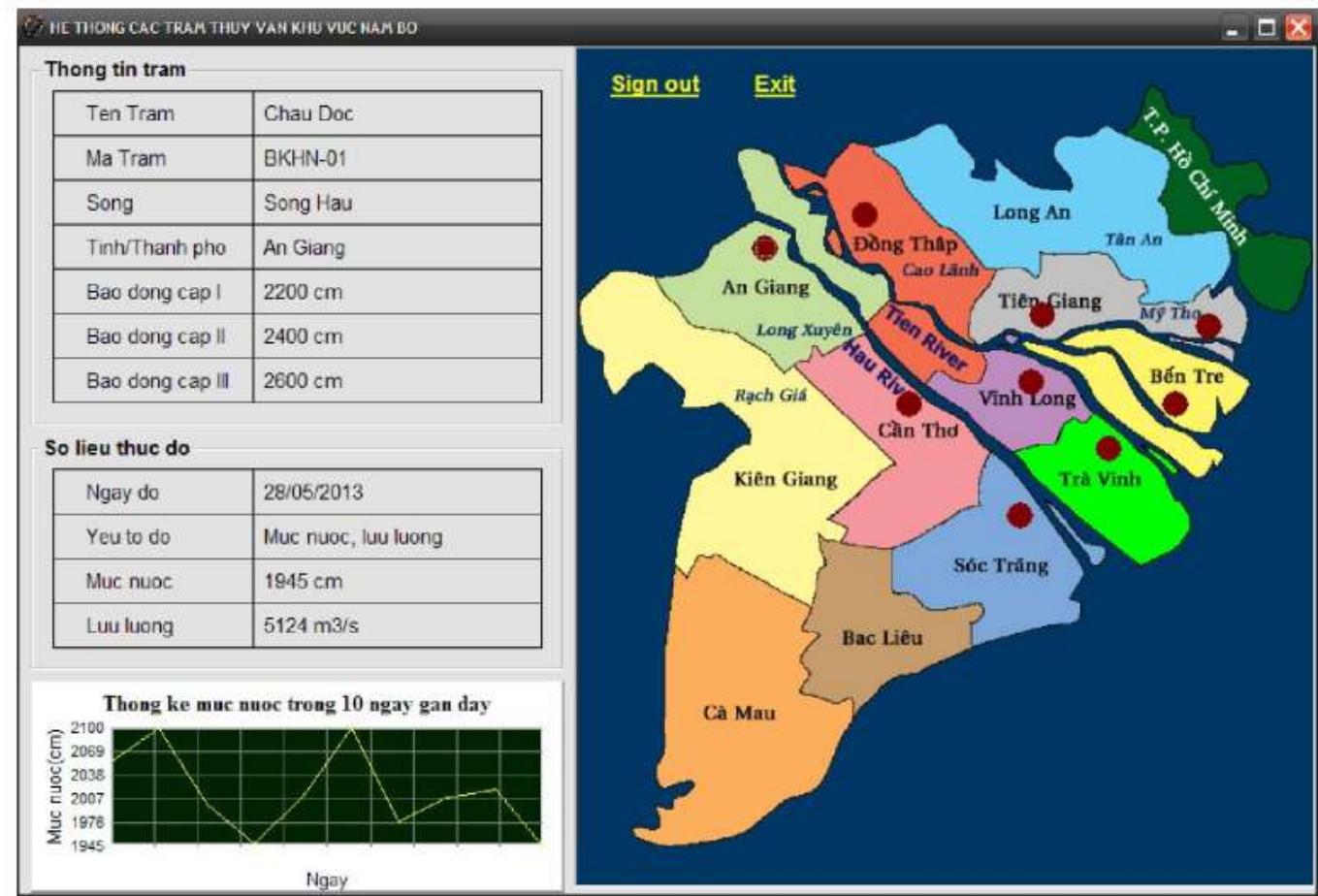
❖ CPU + SIM



Project: Example

Using a microcontroller to simulate SCADA (temperature and water level monitoring)

❖ UI



Project: Example

Using a microcontroller to simulate SCADA (temperature and water level monitoring)

❖ Get information via sms



Requirements

- ☐ Students are required to attend classes and complete all assigned exercises and reports.
- ☐ Excessive talking in class that disrupts the instructor or the classroom environment is not allowed; violations will result in deductions from the attendance score.
- ☐ During quizzes and examinations, academic dishonesty such as cheating or copying others' work is strictly prohibited.
- ☐ Requests for grade adjustments and late submissions after the deadline are not accepted.
- ☐ Any violation of these rules will result in a score of zero for all related assessments.